

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An autonomous Institution affiliated to Anna University)

Degree & Branch	BE Computer Science & Engineering	Semester	V
Subject Code & Name	UCS3512 & Principles of Machine Learning Laboratory		
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Experiment 3: Classification of MNIST Handwritten Digits using Support Vector Machine(SVM)

OBJECTIVE

To implement a Support Vector Machine (SVM) classifier for recognizing handwritten digits from the MNIST dataset. Perform image preprocessing and exploratory data analysis, compare different kernel functions, visualize classification results, and analyze the performance of classifier using suitable evaluation metrics.[CO1, K3].

Dataset

The MNIST Handwritten Digits Dataset consists of 70,000 grayscale images of handwritten digits (0-9)

- Total images: 70,000
- Training images: 60,000
- Testing Images: **10,000**
- Image Size: **28 × 28 pixels**
- Number of Classes: **10**

Each image is represented as a grayscale matrix containing handwritten numerical digits.

Dataset Reference

UCI Repository:

<https://archive.ics.uci.edu/dataset/683/mnist+database+of+handwritten+digits>

Code for import the dataset using TensorFlow Dataset:

```
from tensorflow.keras import datasets
(x_train, y_train), (x_test, y_test) = datasets.mnist.load_data()
```

Brief Theory (For Lab Understanding)

Support Vector Machine (SVM)

Support Vector Machine is a supervised machine learning algorithm used for classification and regression. It constructs an optimal separating hyperplane that maximizes the margin between different classes.

Kernel Functions:

Linear Kernel

Suitable for linearly separable data.

$$K(x_i, x_j) = x_i^T x_j$$

Polynomial Kernel

Maps data into higher-dimensional feature space.

$$K(x_i, x_j) = (x_i^T x_j + c)^d$$

Radial Basis Function (RBF) Kernel

Most commonly used nonlinear kernel.

$$K(x_i, x_j) = e^{-\gamma \|x_i - x_j\|^2}$$

Sigmoid Kernel

Inspired by neural networks.

$$K(x_i, x_j) = \tanh(\gamma x_i^T x_j + c)$$

Image Preprocessing: Appropriate preprocessing using scikit-image (skimage) should be performed.

Implementation Steps

1. Load the MNIST dataset.
2. Visualize sample handwritten digit images
3. Normalize image pixel values
4. Apply suitable image preprocessing techniques:
 - a. Image enhancement
 - b. Gaussian smoothing
 - c. Median filtering
 - d. Denoising
 - e. Histogram equalization
 - f. Feature extraction
5. Flatten each image into a feature vector.
6. Split the dataset into training and testing sets (if not already separated)
7. Train SVM classifiers using:
 - a. Linear kernel
 - b. Polynomial kernel
 - c. RBF kernel
 - d. Sigmoid kernel
8. Predict the digit labels
9. Evaluating model performance

10. Compare the performance of different kernels

Required Visualizations (During Coding)

Students must generate and include:

- Sample MNIST digit images
- Histogram of image intensities and equalization results
- Original vs. processed images
- Feature distribution plots
- Confusion matrix
- Classification report
- Accuracy comparison bar chart
- Kernel comparison graph
- Misclassified digit images

Performance Metrics to be Reported

- Training accuracy
- Testing accuracy
- Precision
- Recall
- F1-score
- Confusion matrix
- Classification report
- Training time
- Prediction time

Hyperparameter Search Space

- Kernel {Linear, polynomial, RBF, sigmoid}
- C = {0.1, 1, 10, 100}
- Gamma = {scale, auto, 0.001, 0.01, 0.1}
- Degree = {2, 3, 4} (polynomial only)

Hyperparameter Tuning Results

Table 1: Hyperparameter Tuning Summary

Kernel	Best C	Best Gamma	Best Degree	Best Cross Validation Accuracy
Linear			NA	
Polynomial				
RBF			NA	
Sigmoid			NA	

Cross-Validation Performance (K = 5)

Table 2: Cross-Validation Performance

Kernel	Accuracy	Precision	Recall	F1 Score
Linear				
Polynomial				
RBF				
Sigmoid				

Table 3: Test Set Performance

Kernel	Training Accuracy	Testing Accuracy	Precision	Recall	F1 Score
Linear					
Polynomial					
RBF					
Sigmoid					

Test Set Performance

Table 4: Coefficient Comparison

Kernel	Training Accuracy	Testing Accuracy	Precision	Recall	F1 Score
Linear					
Polynomial					
RBF					
Sigmoid					

Conclusion

Summarize the performance of all SVM kernel functions, identify the optimal kernel and hyperparameters, discuss the impact of preprocessing on classification accuracy, and justify the most suitable SVM model for handwritten digit recognition.

Report Format (Mandatory)

1. Aim and Objective
2. Dataset Description
3. Image Preprocessing
4. Exploratory Data Analysis
5. SVM Model Development
6. Hyperparameter Tuning
7. Visualizations

8. Performance Tables
9. Kernel Comparison
10. Analysis and Inference
11. Observations
12. Conclusion
13. References

References

- Scikit-learn Documentation – Support Vector Machines
<https://scikit-learn.org/stable/modules/svm.html>
- TensorFlow Keras MNIST Dataset Documentation
https://www.tensorflow.org/api_docs/python/tf/keras/datasets/mnist
- UCI Machine Learning Repository – MNIST Handwritten Digits
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- MNIST Database by Yann LeCun
<http://yann.lecun.com/exdb/mnist/>
- Scikit-image Documentation
<https://scikit-image.org/>
- OpenCV Histogram Equalization Tutorial
https://docs.opencv.org/4.x/d5/daf/tutorial_py_histogram_equalization.html
- Scikit-learn Model Selection Documentation
https://scikit-learn.org/stable/modules/grid_search.html

Additional Questions:

1. Find the optimal hyperplane using SVM Kernels for AND gate.